

VALERIYA SLOVIKOVSKAYA

Lead Software Architect — AI, Agentic & Data Systems

Software architect, 15+ years in production systems — almost entirely startups and internal startups — across fintech, e-commerce, pharma and publishing. I re-architect fragile or failing production services so they stay up, and design and build the ML and data systems the business runs on. I lead when there is a team to lead, and deliver substantial work alone when there isn't. PhD in physics.

Looking for: agentic AI/LLM engineering · LLM evaluation and observability · AI integration · software architecture and platform reliability.

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Experience

Lead Software Architect — AI & Data Systems

HomeServe (UK) — HomeServe Finance (internal fintech startup), then HomeServe Ltd

Apr 2024 — Present

Consumer-credit platform — engineering lead, Loan Management & Payments:

principal contributor to the loan-servicing and collection core of a regulated UK consumer-credit business launched from zero — Django services on AWS Lambda and Terraform, event-driven integration (SQS/EventBridge), direct-debit collections, loan servicing, arrears and fee logic, and early-settlement calculations implemented against UK consumer-credit legislation. Implemented direct-debit integrations with both Stripe and GoCardless, through to decommissioning the superseded payment service. Grew the team from one to seven in ten months, onboarding or training eleven engineers — several reassigned elsewhere once productive. Moved the core lending service from single-author to mandatory reciprocal review, designed the end-to-end API test framework and taught a team with no prior experience to build it, and wired test failures into alerting so no non-green build reached production.

Production incident response: recurring first responder across payment and collection incidents, and re-architect for services released in a fragile state — on the largest, isolated the root cause across the affected book, flagged the consumer-fairness consequence before proposing a fix, and coordinated remediation.

ML & data — delivered independently: built a LightGBM customer-acquisition model over a

38M-row customer base — 0.77 test AUC, 8× lift in the top 1% and 4.2× in the top decile — trained on a 100k stratified sample for one-minute retraining, with a 533-run hyperparameter search tracked in Weights & Biases. Scored 2.9M households for a live direct-mail campaign and documented the score cliff showing model signal was confined to the top ~3%, correctly predicting the remaining waves would track population baseline. Also: an in-house probability-of-repayment model on credit-bureau data; census-to-feature ETL across five countries with checksummed sources and documented lineage; satellite-imagery solar-panel detection (computer vision); a k-anonymity/re-identification-risk PII-governance framework used as shared infrastructure; and a field application for capturing customer visit notes. Directs AI coding agents (Claude Code) daily; extending this personally into agentic/LLM engineering.

Model validation and audit: found and fixed a validation-set misuse defect in my own training pipeline (early stopping on the test split), quantified its effect on every published figure — the contamination had flipped the sign of a feature-ablation conclusion — and added a pure-noise regression test pinning the corrected path to chance. Separately audited a parallel production pipeline, finding household-ID leakage, unstratified splits and a broken age derivation behind near-random holdout performance.

Senior Software Engineer FairScore (London)

Aug 2022 — Apr 2024

Product: Updraft Credit

Built the core loan-servicing and payment-collection engine for a consumer lender with a ~50,000-loan book, replacing a third-party microfinance platform that could not represent the legislated cases — a year of work ending in

its decommissioning. The base case is trivial; the engine exists for the abnormal paths, each of which must stay legislation-compliant — missed and late payments, forbearance, repayment-day changes, write-offs, early settlement ahead of the calculated date, partial refunds, and statutory breathing space. Implemented its regulatory reporting — Bank of England returns and ESMA securitisation reporting — behind a test suite larger than the production code. The same core served both Updraft's own loans and the HVAC-financing line HomeServe commissioned, which HomeServe later brought in-house — carrying the codebase across with it. Stopped a sustained SMS-pumping attack costing more in messaging than genuine traffic. Rotating IPs and varying timings defeated static rules, so I built a dynamic attack-tracking service driving AWS WAF rules via Cognito triggers and Lambda — the real problem being where to set the threshold, since over-blocking locks legitimate customers out of registration. Root cause was architectural: SMS rather than email registration makes the registration and code-request paths attacker-payable by design. (Python, Django, AWS Serverless, Cognito, WAF, PostgreSQL, MySQL, Terraform)

Senior Enterprise Engineer

A-Tono S.r.l.

Dec 2020 — Aug 2022

Built payment services for a PSD2-regulated payment institution — POS and consumer flows on an event-driven microservice architecture, with PagoPA and Italian interbank integrations. (Java, Spring, Quarkus, PostgreSQL, RabbitMQ, Kafka, Elasticsearch)

Earlier experience (2011–2020)

NTT DATA Italia (Pisa, 2018–2020), clients Chiesi Farmaceutici and Inpeco — built a business-automation (CRM) platform from scratch; parallelised medical-sample processing with RabbitMQ and containerised it, removing scalability bottlenecks. **Zanichelli** (Pisa, 2015–2018) — payments, subscriptions and single sign-on for Italy's leading educational publisher, including the state teacher-voucher scheme. **Home24** (Berlin, 2014–2015) — secure re-architecture of shopping cart and payment flow backed by 100% unit and acceptance test coverage; built the test automation framework. **OneFootball** (Berlin, 2013–2014) — mobile REST API back-end.

Professional.ru (Moscow, 2011–2013) — job-search portal, 2M → 5M users in two years on a platform rebuilt in-house by 25 engineers. Built the targeted-advertising and billing modules.

Skills

— **AI/LLM:** agentic & tool-calling systems, embeddings and semantic similarity search, Anthropic API, Ollama (local/private deployment), Claude Code (daily professional use directing coding agents)

— **ML & data:** scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch, Hugging Face Transformers, SHAP, optbinning, imbalanced-learn; polars, duckdb, pyarrow, dbt, AWS Athena and Glue, geopandas, rasterio; Weights & Biases; leakage analysis and out-of-sample validation discipline

— **Cloud & DevOps:** AWS (Lambda, SageMaker, Athena, Glue, EventBridge, Aurora, RDS, Cognito, WAF, CloudWatch,

Secrets Manager), Azure, Google Cloud; serverless architectures; Terraform, Docker, GitHub Actions, GitLab CI/CD; automated testing and infrastructure automation

— **Languages & stores:** Python (primary — Django, Flask), SQL, BASH; earlier production experience in Java (Spring, Quarkus), GoLang and JavaScript/TypeScript. PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch

— **Architecture & governance:** event-driven integration (SQS, EventBridge), microservices, distributed systems, design patterns; k-anonymity, re-identification-risk analysis, data minimisation

Education & awards

Pisa University (Italy) — Laurea magistrale in Informatica Umanistica, 110/110 — distributed and parallel systems, computational linguistics, NLP, information retrieval, machine learning.

Urals State University (Russia) — PhD (Candidate of Sciences), Theoretical Physics; 5-year Specialist degree in Physics, thesis in solid state physics; Master in Art History.

Codility — Golden Award (Palladium, 2020).

Independent work

— LLM agent over Italian tax law: extracts structured input from unstructured regulation and orchestrates a deterministic calculation engine — every result traceable to a formula and its legal source, with a mandatory human-verification gate before anything is acted on.

— Local-embeddings signal engine: mines text sources and ranks them by cosine similarity against a target description (Ollama, fully local).

— Agent-observability dashboard: cost, tokens, tool calls and per-session efficiency scoring.

— Error-mining pipeline: extracts recurring grammatical errors from my English and Italian writing into a corrections dashboard.

Publications

Transfer Learning from Transformers to Fake News Challenge Stance Detection (FNC-1). LREC 2020, Marseille — degree thesis; state-of-the-art on FNC-1 by fine-tuning BERT, RoBERTa and XLNet. (<http://www.lrec-conf.org/proceedings/lrec2020/pdf/2020.lrec-1.152.pdf>)

Languages: Russian (native); English, Italian, French, German, Spanish (fluent).